

Engy Saeed Abdelhamed Ali

Data Analyst

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Summary

Strategic Data Analyst & Machine Learning Practitioner with a Bachelor of Data Science. Expert in transforming high-volume datasets into growth-oriented insights across **Retail, Healthcare, and Finance**. Proven track record of leveraging advanced statistical modeling and clustering to drive **30%+ profit increases** and optimize operational workflows. Skilled at bridging technical complexity with executive-level strategy to deliver high-impact, data-driven solutions.

Technical Skills

- **Database Management (SQL/SSMS):** Complex Query Development, Multi-table Joins, Data Aggregations, Data Manipulation (DML), and Schema Management.
- **Data Analysis & Statistics:** Exploratory Data Analysis (EDA), Statistical Testing (A/B Testing, Hypothesis Testing), ETL Processes, Data Cleaning, KPI Monitoring, and Web Scraping.
- **Languages:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R Programming (Statistical Analysis, Machine Learning).
- **Data Visualization:** Power BI (DAX, Power Query, Data Modeling, KPI Tracking, Interactive Dashboards), Excel (Advanced Formulas, Pivot Tables, Data Modeling, Dashboards).
- **Machine Learning:** Supervised & Unsupervised Learning, Clustering, Predictive Modeling, Correlation Analysis, Feature Selection.
- **Domain Expertise:** Sales & Retail Analytics, Healthcare Operations, HR Analytics, Call Center Performance, Logistics & Sustainability.

Soft Skills

- Expertise in strategic problem-solving, meticulous attention to detail, and presenting technical insights to drive executive-level business decisions.
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Experience

Freelance Data Analyst | Remote | Jun 2025 – Dec 2025

- **Optimized** commercial operations for diverse clients by delivering data-driven insights that contributed to a 12% increase in identified high-margin opportunities.
- **Architected** automated HR Performance Ecosystems and Sales trackers, reducing manual data processing and reporting time by 20+ hours per month.
- **Developed** interactive dashboards and diagnostic tools that enabled stakeholders to visualize hidden market trends and proactively mitigate operational risks.

Data Analyst Intern | EYouth | Remote | Jun 2025 – Jul 2025

- **Extracted** and validated large-scale sales datasets from SQL Server using complex queries, ensuring 100% data integrity for regional performance reporting.
- **Analyzed** cross-functional sales activity to identify key growth drivers, providing the diagnostic data used by management to support strategic expansion initiatives.
- **Streamlined** business intelligence workflows by designing unified Power BI solutions, reducing the time required for weekly executive data reviews by 30%.

Data Analyst Intern | DEPI | Alexandria, Egypt | Apr 2024 – Oct 2024

- **Engineered** an end-to-end business intelligence solution for an airline loyalty program, managing a comprehensive database of 5,000+ customers to track engagement.
- **Executed** advanced customer segmentation (Clustering) and retention analysis, identifying high-value passenger demographics to improve loyalty program targeting.
- **Designed** a suite of interactive KPI dashboards monitoring points velocity and service efficiency, providing the clarity required for multi-departmental strategy shifts.

Machine Learning Intern | My Communication | Alexandria, Egypt | Jul 2023 – Sep 2023

- **Developed** a predictive modeling engine for real estate valuation, utilizing Python to deliver reliable price estimates in a volatile Egyptian housing market.
 - **Implemented** robust feature engineering and statistical validation protocols, enhancing model reliability through rigorous hyperparameter optimization and data cleaning.
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Key Projects

Customer Segmentation & Risk Analysis | Python (Scikit-Learn), Power BI

- **Operationalized** an Unsupervised Machine Learning model (K-Means) to cluster 5,000+ customers based on RFMT/CFMT metrics, identifying distinct behavioral groups.
- **Pinpointed** high-revenue segments with elevated cancellation risks, providing a diagnostic view of "At-Risk" value that was previously hidden in raw data.
- **Developed** a strategic dashboard that translated complex clustering results into actionable retention and marketing campaigns for executive stakeholders.

Hospital ER Operational Intelligence | Power BI, DAX, Data Modeling

- **Optimized** clinical resource allocation by analyzing 9,000+ patient encounters to identify peak workload patterns (71% weekday concentration) and throughput bottlenecks.
- **Quantified** operational efficiency through custom DAX measures, tracking a 35-minute average wait time and correlating it with patient satisfaction scores.
- **Delivered** data-backed scheduling recommendations to reduce wait times and improve staff utilization during high-volume walk-in periods.

Adventure Works Market Intelligence | Excel (Power Pivot), Power Query

- **Architected** a comprehensive sales ecosystem identifying that 41% of total profit was driven by the 50+ age demographic, specifically within the US and Australian markets.
- **Isolated** seasonal trends showing Q2 as the primary revenue driver (31% of annual profit), enabling more accurate inventory and marketing budget forecasting.
- **Automated** YoY performance tracking for revenue and profit KPIs, reducing manual reporting overhead and ensuring real-time visibility into product pricing effectiveness.

HR Performance & Talent Diagnostics | Power BI, DAX, Advanced Analytics

- **Engineered** a talent intelligence dashboard to correlate educational backgrounds with high-performance metrics across diverse organizational departments.
 - **Identified** key drivers of employee satisfaction and retention, providing a diagnostic tool for HR leadership to mitigate turnover risks.
 - **Streamlined** the visualization of cross-functional performance data, allowing for faster identification of high-potential talent within the organization.
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Education:

Bachelor of Data Science

Faculty of Computers and Data Science, Alexandria University

Jul 2025 | Grade: Very Good

Certificates:

- **Artificial Intelligence Internship** – My Communication | Jul 2023 – Sep 2023
 - **Microsoft Power BI Engineer** – DEPI | Apr 2024 – Oct 2024
 - **Internet of Things (IoT)** – ITI | Aug 2024 – Aug 2024
 - **Artificial Intelligence** – NTI (Score: 92%) | Apr 2025 – May 2025
 - **Data Analysis** – EYouth | Jun 2025 – Jul 2025
 - **HR Member** – HackerRank Campus Club (FCDS) | Jan 2023 – Dec 2023
 - **HR Member** – PES (Programmers Elite School) | Jan 2023 – Dec 2023
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Languages:

- **English:** Fluent
- **Arabic:** Native